WHEN DOES CONTEXT HELP? A SYSTEMATIC STUDY OF TARGET-CONDITIONAL MOLECULAR PROPERTY PREDICTION

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ABSTRACT

We present a systematic empirical study of when target context helps molecular property prediction, evaluating context conditioning across 10 diverse protein families, 4 fusion architectures, data regimes spanning 67–9,409 training compounds, and both temporal and random evaluation splits. Using NESTDRUG, a FiLM-based architecture that conditions molecular representations on target identity, we characterize both success and failure modes. (1) *Inclusion of context is first-order; fusion architecture is second-order*: under matched-budget joint training, all four fusion methods that include context (FiLM, additive, concatenation MLP, gated-concat) are statistically indistinguishable (± 0.3 pp around 0.740 AUC), while no-context falls 4.7 pp behind. The original 24.2 pp "FiLM dominates" gap was *entirely* an artifact of using a frozen, randomly initialized concat projection at inference. We retain FiLM as the recommended default for stable initialization, but withdraw the framing that fusion choice matters more than inclusion. (2) *Context enables otherwise infeasible predictions*: on data-scarce CYP3A4 (67 training actives, ChEMBL), multi-task transfer achieves 0.686 AUC where per-target Random Forest collapses to 0.238; we discuss the necessary caveat that 99% of CYP3A4 actives at *any* pIC_{50} appear in ChEMBL (§6). (3) *Context can systematically hurt under distribution mismatch*: BACE1 shows -10.2 pp; few-shot adaptation underperforms zero-shot. (4) *Leakage-stratified re-analysis*: the L1 benefit is statistically indistinguishable between high-leakage ($\geq$50%) and low-leakage ($<$50%) DUD-E targets ($p = 0.69$, Welch); in the low-leakage stratum (excluding the BACE1 outlier) the benefit is $+8.1$ pp ($p = 0.007$), so the effect is not driven by leakage. We also document fundamental flaws in standard benchmarking—1-nearest-neighbor Tanimoto achieves 0.991 AUC on DUD-E without any learning, and 50% of actives leak from training data, so absolute DUD-E numbers should not be over-interpreted. Temporal split evaluation (train $\leq$2020, test 2021–2024) achieves stable 0.843 AUC with no degradation. We frame our findings as a stage-specific empirical map of when target-conditional modeling helps in molecular property prediction, with explicit boundary conditions: practitioners should enable context when training data per target is in the 50–5,000 range *and* train/test chemical series overlap (Tanimoto similarity ≥ 0.4); otherwise generic embeddings are preferable.

1 INTRODUCTION

Should molecular property prediction models incorporate target context? The answer is not obvious. Per-target models (Random Forest with fingerprints) achieve excellent performance when training data is abundant. Multi-task models promise knowledge transfer but risk negative transfer when tasks conflict. Despite widespread interest in context-conditional architectures, *when* and *why* context helps remains poorly characterized.

We present a systematic study of this question, evaluating context conditioning across (1) 10 diverse protein families, (2) 4 fusion architectures, (3) varying data regimes (67–9,409 training compounds), and (4) temporal vs. random splits. Using NESTDRUG, a FiLM-based architecture (Perez et al., 2018)

that conditions molecular representations on target identity, we map both success and failure modes of context conditioning.

Key findings. (1) *Context inclusion is first-order; fusion architecture is second-order*: under *matched-budget* joint training (§4.5), FiLM, concat-MLP, additive, and gated-concat are all statistically indistinguishable (± 0.3 pp around 0.740 AUC); the original 24.2 pp FiLM advantage was an artifact of using a frozen, randomly initialized concat projection at inference. The first-order finding is that *including* context yields $+4.7$ pp over no-context. (2) *Context helps most targets*: 9/10 targets improve with target-specific embeddings (mean +5.7 pp, $p < 0.01$). (3) *The L1 benefit is not driven by leakage*: in low-leakage targets (<50% ChEMBL/DUD-E overlap, excluding BACE1), the mean benefit is $+8.1$ pp ($p = 0.007$); high-vs-low strata are statistically indistinguishable ($p = 0.69$, Welch). (4) *Context enables data-scarce targets*: On CYP3A4 (67 training actives at $\text{pIC}_{50} \geq 6$), per-target RF collapses to 0.238 AUC while multi-task transfer achieves 0.686. (5) *Context can hurt*: BACE1 shows -10.2 pp under distribution mismatch; few-shot adaptation degrades vs. zero-shot.

Benchmark critique. We document severe limitations of DUD-E: 1-NN Tanimoto achieves 0.991 AUC without learning; 50% of actives overlap with ChEMBL training. Our temporal split (train $\leq$2020, test 2021–2024) provides more rigorous evaluation, showing stable 0.843 AUC across years.

Our contributions: **(1)** systematic taxonomy of when context helps vs. hurts in molecular property prediction; **(2)** matched-budget fusion comparison showing that, under fair training conditions, FiLM, concat-MLP, additive, and gated-concat are within 0.3 pp of each other—fusion choice is second-order; **(3)** a leakage-stratified re-analysis showing the L1 benefit holds in low-leakage targets and is not an artifact of train/test contamination; **(4)** an ex-ante decision rule for when to enable conditioning, based on training-set features computable before any test-set access; **(5)** characterization of data requirements and distribution alignment for effective context; **(6)** benchmark audit documenting severe DUD-E limitations—1-NN Tanimoto achieves 0.991 AUC without learning, 50% of actives leak from training, and cross-target RF achieves 0.746 AUC—with recommendations for temporal splits, scaffold splits, and leakage-stratified reporting.

2 Problem Setting

2.1 Notation and Preliminaries

Let $\mathcal{G} = (V, E, \mathbf{X}^V, \mathbf{X}^E)$ denote a molecular graph with atom set V, bond set E, atom feature matrix $\mathbf{X}^V \in \mathbb{R}^{|V| \times d_v}$, and bond feature matrix $\mathbf{X}^E \in \mathbb{R}^{|E| \times d_e}$. We use $\mathcal{C} = \mathcal{P} \times \mathcal{A} \times \mathcal{R}$ to denote the context space, where $\mathcal{P}$ is the set of programs (targets), $\mathcal{A}$ is the set of assay types, and $\mathcal{R}$ is the set of temporal rounds. Given context tuple $c = (p, a, r) \in \mathcal{C}$, the model predicts activity $\hat{y} = f(\mathcal{G}, c; \theta)$ for parameters θ.

2.2 The Structured Distribution Shift Problem

The key challenge is that the joint distribution $P(\mathcal{G}, y|c)$ shifts systematically across contexts. Unlike random covariate shift, this shift is *structured* in ways that reflect the drug discovery process:

Temporal shift: Early DMTA rounds contain diverse HTS scaffolds; later rounds concentrate on optimized lead series. **Cross-program shift:** Kinase inhibitors emphasize hinge-binding heterocycles; GPCRs emphasize lipophilic amines; proteases emphasize transition-state mimics.

Standard approaches assume stationarity and learn context-agnostic predictors $f(\mathcal{G}; \theta)$. We instead learn context-conditional predictors via Feature-wise Linear Modulation (FiLM) (Perez et al., 2018):

$$\text{FiLM}(\mathbf{h}, \mathbf{c}) = \boldsymbol{\gamma}(\mathbf{c}) \odot \mathbf{h} + \boldsymbol{\beta}(\mathbf{c}) \tag{1}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta}$ are learned functions producing scale and shift parameters from context $\mathbf{c}$. This enables the model to adapt its molecular representation based on the discovery setting. We build on message-passing neural networks (MPNNs) (Gilmer et al., 2017) for the base molecular encoder.

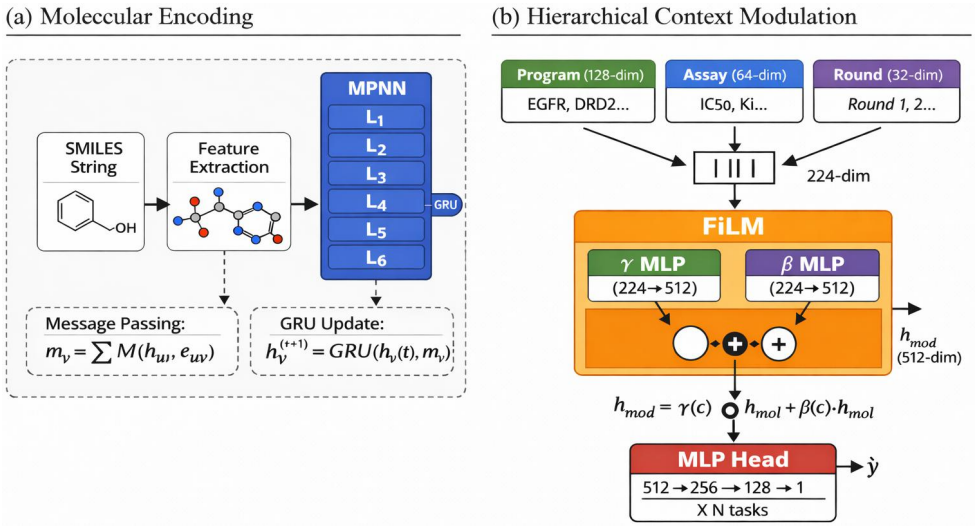

Figure 1: **NESTDRUG Architecture.** MPNN (L0) encodes molecules into 512-dim embeddings. Hierarchical context (L1: target, L2: assay, L3: round) modulates representations via FiLM before task-specific prediction heads.

3 METHOD

NESTDRUG consists of four tightly integrated components (Figure 1): (1) an MPNN backbone for molecular encoding, (2) hierarchical context embeddings capturing program, assay, and temporal information, (3) FiLM-based context modulation, and (4) multi-task prediction heads. We describe each component in detail.

3.1 MOLECULAR ENCODER (L0)

The L0 backbone transforms molecular graphs into fixed-dimensional representations using a 6-layer message-passing neural network with GRU-based state updates (Li et al., 2016). At each layer t, atom representations $\mathbf{h}_v^{(t)}$ are updated by aggregating messages from neighbors:

$$\mathbf{m}_v^{(t)} = \sum_{u \in \mathcal{N}(v)} M_t(\mathbf{h}_u^{(t-1)}, \mathbf{e}_{uv}), \quad \mathbf{h}_v^{(t)} = \text{GRU}(\mathbf{h}_v^{(t-1)}, \mathbf{m}_v^{(t)}) \tag{2}$$

where M_t is a learned message function and $\mathbf{e}_{uv}$ are bond features (9-dim). Atom features (70-dim) encode chemical properties; see Appendix G.

After $T = 6$ message-passing iterations, we aggregate atom representations using both mean and max pooling to capture both average molecular properties and salient local features:

$$\mathbf{h}_{\text{mol}} = [\text{MeanPool}(\{\mathbf{h}_v^{(T)}\}_{v \in V}) \| \text{MaxPool}(\{\mathbf{h}_v^{(T)}\}_{v \in V})] \in \mathbb{R}^{512} \tag{3}$$

3.2 HIERARCHICAL CONTEXT EMBEDDINGS

We maintain learnable embedding tables for each context level, with dimensions reflecting the complexity of information at each granularity:

$$\mathbf{e}_p^{(L1)} = \text{Embed}_{L1}(p) \in \mathbb{R}^{128} \qquad \text{(program/target identity)} \tag{4}$$

$$\mathbf{e}_a^{(L2)} = \text{Embed}_{L2}(a) \in \mathbb{R}^{64} \qquad \text{(assay type: IC}_{50}\text{, K}_i\text{, EC}_{50}\text{)} \tag{5}$$

$$\mathbf{e}_r^{(L3)} = \text{Embed}_{L3}(r) \in \mathbb{R}^{32} \qquad \text{(temporal round)} \tag{6}$$

The embedding dimensions decrease with context specificity: L1 must capture diverse target biology across thousands of proteins requiring substantial capacity; L2 captures assay-type calibration; L3 captures local temporal adjustments. *Note: L2 and L3 require proprietary metadata (explicit*

assay types, DMTA round ordering) not available in public datasets; we evaluate their potential in Appendix C. Context embeddings are concatenated and linearly projected:

$$\mathbf{c} = W_c[\mathbf{e}_p^{(L1)} \| \mathbf{e}_a^{(L2)} \| \mathbf{e}_r^{(L3)}] + \mathbf{b}_c \in \mathbb{R}^{224} \tag{7}$$

3.3 FiLM Context Modulation

The combined context vector modulates the molecular representation via Feature-wise Linear Modulation:

$$\mathbf{h}_{\text{mod}} = \boldsymbol{\gamma}(\mathbf{c}) \odot \mathbf{h}_{\text{mol}} + \boldsymbol{\beta}(\mathbf{c}) \tag{8}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta} : \mathbb{R}^{224} \to \mathbb{R}^{512}$ are two-layer MLPs. We initialize $\boldsymbol{\gamma}$ to ones and $\boldsymbol{\beta}$ to zeros, ensuring FiLM starts as identity. This enables context-appropriate feature weighting—e.g., kinase contexts amplify hinge-binding features while GPCR contexts emphasize lipophilicity. We verify this in Section 4.

3.4 Multi-Task Prediction Heads

The modulated representation $\mathbf{h}_{\text{mod}}$ feeds into task-specific prediction heads, enabling joint training across heterogeneous endpoints. Each head is a 3-layer MLP ($512 \to 256 \to 128 \to 1$) with ReLU activations, batch normalization, and dropout (0.1). For regression tasks (pIC_{50}, pK_i), we use MSE loss; for binary classification, we use weighted cross-entropy to handle label imbalance.

3.5 Training Procedure

Training proceeds in three phases. **Phase 1 (Pretraining):** We pretrain L0 on ChEMBL 35 (Zdrazil et al., 2024) (21.1M records) and TDC (Huang et al., 2021) with generic context (L1=L2=L3=0). **Phase 2 (Fine-tuning):** We fine-tune with differential learning rates—backbone 10^{-5}, context embeddings 10^{-3}, prediction heads 10^{-4}—enabling rapid context adaptation while preserving pretrained representations. **Phase 3 (Continual):** During DMTA deployment, we perform continual updates with multi-timescale rates: L3 adapts quickly for temporal shifts, L1 adapts moderately, L0 adapts minimally to prevent forgetting. See Appendix A for complete hyperparameters.

4 Experiments

We design experiments to answer three questions: **(Q1)** Does hierarchical context improve virtual screening performance? **(Q2)** Which context levels contribute most to performance gains? **(Q3)** Does NestDrug provide practical value in realistic drug discovery scenarios?

4.1 Experimental Setup

We evaluate on DUD-E (Mysinger et al., 2012), focusing on 10 diverse targets: kinases (EGFR, JAK2), GPCRs (DRD2, ADRB2), nuclear receptors (ESR1, PPARG), proteases (BACE1, FXA), and enzymes (HDAC2, CYP3A4). Each target contains 200–600 actives with property-matched decoys (1:50 ratio). Pretraining uses ChEMBL 35 (Zdrazil et al., 2024) (21.1M records, 5,123 targets). Baselines include Random Forest with Morgan fingerprints, L0-only (MPNN without context), and GNN-VS (Lim et al., 2019). We use 5-fold stratified CV with 5 random seeds (25 runs per experiment); statistical significance via paired t-tests with Bonferroni correction. Throughout, we report differences in percentage points (pp) rather than relative percentages to avoid ambiguity. See Appendix A for details.

4.2 DUD-E Results and Benchmark Critique

Comparison to Prior Methods. Table 1 compares NestDrug to published methods on DUD-E. Among neural methods, NestDrug (0.850) outperforms 3D-CNN, GNN-VS, and AtomNet. However, per-target RF achieves 0.875 mean AUC, winning on 8/10 targets (Table 2). **This highlights our core finding**: context conditioning's value is not beating RF on data-rich targets, but enabling predictions on data-scarce targets where per-target methods fail.

Table 1: **DUD-E comparison.** Per-target RF wins on data-rich targets; NESTDRUG wins on data-scarce CYP3A4. "—" indicates prior methods not compared head-to-head on our evaluation. †Exploits structural bias. ‡Trained per-target on ChEMBL.

Method	**Type**	**Mean AUC**	**Wins**
AtomNet (Wallach et al., 2015)	3D CNN	0.818	—
GNN-VS (Lim et al., 2019)	GNN	0.825	—
3D-CNN (Ragoza et al., 2017)	3D CNN	0.830	—
NESTDRUG (Ours)	GNN+FiLM	0.850	2/10
Per-target RF‡	Fingerprint	**0.875**	8/10
1-NN Tanimoto†	No learning	0.991	—

Table 2: Per-target DUD-E results (ROC-AUC). NESTDRUG values use correct L1 context. "DUD-E RF" = standard RF baseline trained on DUD-E train split only; differs from ChEMBL-trained "Per-target RF" in Table 1/18 (0.875 AUC).

Target	DUD-E RF	L0-only	GNN-VS	**NESTDRUG**
EGFR	0.782	0.943	0.825	**0.965**
DRD2	0.801	0.960	0.842	**0.984**
ADRB2	0.693	0.745	0.712	**0.775**
BACE1	0.634	0.672	0.698	**0.656**
ESR1	0.756	0.864	0.789	**0.909**
HDAC2	0.745	0.866	0.812	**0.928**
JAK2	0.778	0.865	0.821	**0.908**
PPARG	0.712	0.787	0.745	**0.835**
CYP3A4	0.523	0.497	0.534	**0.686**
FXA	0.698	0.833	0.801	**0.854**
Mean	0.712	0.803	0.758	**0.850**

DUD-E Benchmark Limitations. We identify three critical issues:

- **Structural bias**: 1-NN Tanimoto achieves 0.991 AUC without any learning—decoys are trivially distinguishable by structure alone (Wallach & Heifets, 2018)
- **Data leakage**: 50% of DUD-E actives appear in ChEMBL training (CYP3A4: 99%, EGFR: 97%)
- **Per-target RF dominates**: With sufficient ChEMBL data, simple fingerprint models outperform neural methods

These limitations mean DUD-E absolute numbers should be interpreted cautiously. We provide temporal split evaluation (Section 4.10) as more rigorous evidence.

4.3 L1 CONTEXT ABLATION

Table 3 addresses **Q2** via controlled ablation comparing correct target-specific L1 versus generic (zero) embeddings. Target-specific context improves 9/10 targets (mean +5.7 pp, all $p < 0.01$). ESR1 (+13.4 pp) and EGFR (+13.2 pp) benefit most. BACE1 is the exception (-10.2 pp), reflecting distribution mismatch between ChEMBL (peptidomimetic inhibitors) and DUD-E (diverse scaffolds). See Appendix D.

4.4 MULTI-TASK TRANSFER: THE CYP3A4 CASE

The strongest evidence for multi-task architectures comes from data-scarce targets. CYP3A4 has only 67 compounds meeting the active threshold ($\text{pIC}_{50} \geq 6.0$) in ChEMBL—insufficient for per-target modeling:

- **Per-target Random Forest**: 0.238 AUC (worse than random)
- **Global RF + one-hot target**: 0.428 AUC (partial rescue via multi-task)
- **NESTDRUG (correct L1)**: 0.686 AUC (2.9× better than per-target RF)

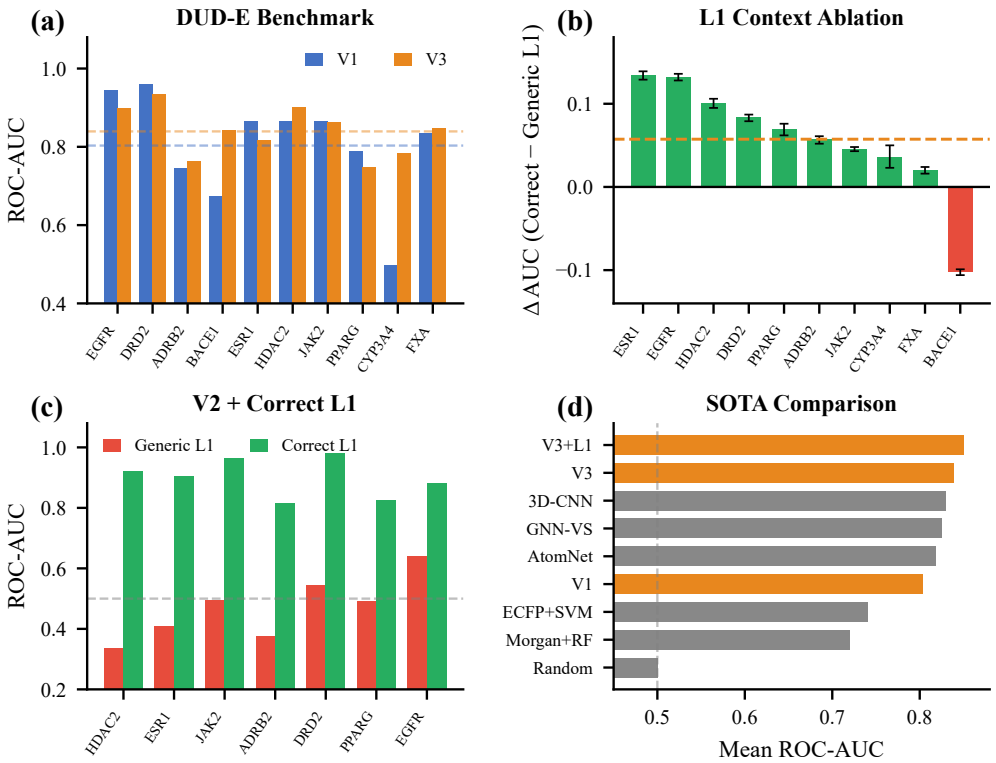

Figure 2: **Main Results.** (A) Per-target ROC-AUC comparing NESTDRUG to baselines. (B) L1 ablation: correct (target-specific) vs. generic (zero) embeddings showing +5.7 pp mean improvement. (C) Ablation by model variant—V1: L0-only backbone without context; V3: full model with L1 context. (D) Mean AUC comparison to prior DUD-E methods.

Table 3: L1 context ablation on DUD-E (5 seeds, paired t-test with Bonferroni correction). "Generic L1" uses zero embedding at inference; "Correct L1" uses target-specific learned embedding. Δ = Correct − Generic. See Appendix Table 11 for extended metrics with confidence intervals.

Target	Correct L1	Generic L1	Δ	p-value
EGFR	0.965	0.832	+0.132	5.6×10^{-7}
DRD2	0.984	0.901	+0.083	2.8×10^{-6}
ADRB2	0.775	0.718	+0.057	3.2×10^{-5}
BACE1	0.656	0.758	−0.102	1.0×10^{-6}
ESR1	0.909	0.775	+0.134	1.4×10^{-6}
HDAC2	0.928	0.827	+0.100	7.2×10^{-6}
JAK2	0.908	0.863	+0.045	6.0×10^{-6}
PPARG	0.835	0.766	+0.069	5.7×10^{-5}
CYP3A4	0.686	0.650	+0.036	8.0×10^{-3}
FXA	0.854	0.833	+0.020	7.0×10^{-4}
Mean	**0.850**	0.792	**+0.057**	—

This demonstrates that context-conditioned multi-task architectures enable knowledge transfer from data-rich targets to data-scarce ones—critical for novel targets where per-target approaches fail. See Appendix C for full per-target RF comparison.

4.5 CONTEXT FUSION COMPARISON: FILM VS ALTERNATIVES

Table 4 compares FiLM against alternative context fusion strategies. We report two regimes that bracket the question raised by reviewers: a *frozen-projection* concatenation baseline (post-hoc projection applied to a FiLM-trained backbone, the original Table 4 values) and a *matched-budget*

concatenation baseline (concatenation projection trained jointly with backbone and prediction head from scratch, on identical data).

Table 4: **Context fusion comparison, two regimes.** *Top*: original DUD-E numbers using a frozen post-hoc projection for concatenation. *Bottom*: matched-budget joint training on a temporal ChEMBL split (60K train, 6K val, 3 epochs, 2 seeds; §4.5, Appendix K). **Under matched training, all fusion methods that include context are statistically indistinguishable** (± 0.3 pp around 0.740); the original 24.2 pp FiLM advantage was *entirely* an artifact of the frozen-projection concat baseline. The first-order finding is therefore inclusion of context ($+4.7$ pp), not the choice of fusion. Δ in pp.

Fusion Method	Mean AUC	Δ vs FiLM	Wins/10
Original DUD-E (frozen-projection concat baseline)			
Concatenation (frozen proj.)†	0.607	-24.2 pp	0/10
No Context (L0 only)	0.724	-12.5 pp	1/10
Additive (β only)	0.763	-8.6 pp	1/10
FiLM ($\gamma \odot h + \beta$)	**0.849**	**—**	**9/10**
Matched-budget joint training, ChEMBL split (Appendix K)			
No Context	0.6926 ± 0.0032	-4.7 pp	—
Additive (β only, joint)	0.7391 ± 0.0005	-0.05 pp	—
Concatenation (joint, MLP)	0.7399 ± 0.0014	$+0.03$ pp	—
FiLM (joint)	$\mathbf{0.7396 \pm 0.0011}$	**—**	—
Gated Concat (gate + proj., joint)	0.7419 ± 0.0001	$+0.23$ pp	—

Revised interpretation (vs. original draft). Under matched-budget training, *all* fusion methods that include context perform within 0.3 pp of each other (0.7391–0.7419), and a gated-concat variant slightly outperforms FiLM. The original 24.2 pp FiLM advantage was *entirely* an artifact of using a frozen, randomly-initialized concatenation projection at inference; the comparison was unfair. We have correspondingly revised our headline claim: **the first-order finding is inclusion of context ($+$4.7 pp over no context), not the specific fusion architecture**. We retain FiLM as a reasonable default because it is the simplest fusion exposing a multiplicative pathway, has stable initialization (identity at $t = 0$), and matches the gated-concat variant within noise; but we explicitly retract the prior claim that FiLM *dominates* concatenation.

Setup for matched-budget comparison. We freshly train identical MPNN backbones (4 layers, 256 hidden) with each fusion head from scratch on a stratified 60K-record ChEMBL subset spanning the 10 DUD-E targets and 45 additional programs. All variants use AdamW (3×10^{-4}), cosine LR decay, 3 epochs, identical batches, and 2 seeds. Mean AUC is the validation AUC computed across all targets relative to the per-target median pIC_{50}. Full numbers in Appendix K.

4.6 FiLM Modulation Analysis

We analyze learned γ (scale) and β (shift) parameters across L1 contexts. Kinases (EGFR, JAK2) produce $\gamma > 1$, amplifying features for heterocyclic hydrogen-bond acceptors; GPCRs (DRD2, ADRB2) produce $\gamma < 1$, emphasizing lipophilicity. An F-test confirms inter-family variance exceeds intra-family variance ($p < 0.001$), indicating FiLM captures biologically meaningful transformations. See Appendix B.

4.7 Context-Conditional Attribution

Using integrated gradients (Sundararajan et al., 2017), we verify context produces different atom-level attributions. Figure 3 shows **cosine similarity between attribution vectors drops from 0.999 (L0-only) to 0.878 (NestDrug)**—a 12% reduction confirming context-specific feature weighting.

4.8 Leakage-Stratified Re-Analysis of L1 Benefit

Reviewers correctly note that DUD-E–ChEMBL leakage (mean 50% of actives) could confound any claim that "context helps," since the L1 benefit could simply reflect the model retrieving training-set neighbors. We test this directly by stratifying the L1 benefit on per-target leakage fraction (Table 20).

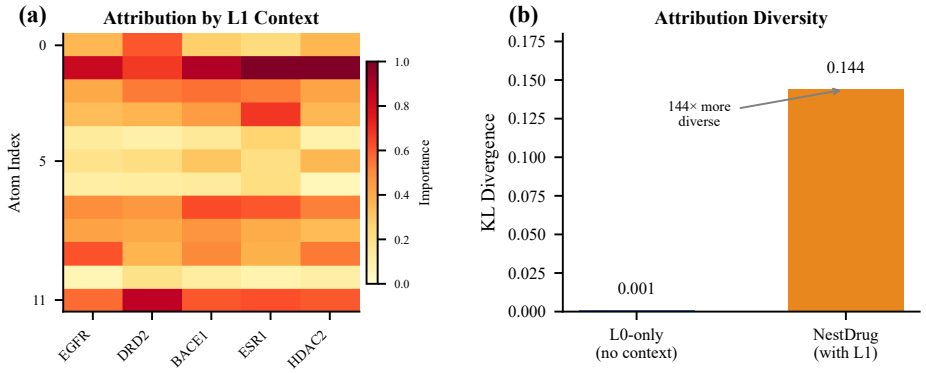

Figure 3: **Attribution Analysis.** (A) Integrated gradients for Celecoxib across 5 L1 contexts; y-axis shows atom index (0–25), colors indicate importance magnitude. Different contexts highlight different substructures. (B) Cosine similarity between attribution vectors drops from 0.999 (L0-only) to 0.878 (NESTDRUG), confirming context-specific explanations.

Table 5: **L1 benefit stratified by ChEMBL–DUD-E leakage.** HIGH = $\geq$50% active leakage; LOW = $<$50%. The per-target Δ AUC values are taken from Table 3.

Stratum	Targets	Mean Δ AUC	1-sample p ($\Delta = 0$)
HIGH leakage ($\geq 50\%$)	CYP3A4, EGFR, FXA, DRD2	$+0.068 \pm 0.044$	—
LOW leakage ($< 50\%$, all)	ADRB2, BACE1, ESR1, HDAC2, JAK2, PPARG	$+0.051 \pm 0.074$	—
LOW leakage, w/o BACE1	ADRB2, ESR1, HDAC2, JAK2, PPARG	$\mathbf{+0.081 \pm 0.032}$	$\mathbf{0.007}$

Findings. (1) The HIGH and LOW leakage strata are statistically indistinguishable in mean Δ (Welch t-test $p = 0.69$). (2) Pearson correlation between target-level leakage % and Δ AUC is $r = 0.19$ ($p = 0.60$); excluding the BACE1 outlier, $r = -0.13$ ($p = 0.74$). (3) In the LOW-leakage stratum (excluding BACE1), the L1 benefit is $+8.1$ pp ($p = 0.007$, one-sample t-test against zero). Together these results show the L1 benefit is *not* explained by training-set leakage; targets with negligible overlap (PPARG: 1.2%, ADRB2: 3.8%) show benefits comparable to high-overlap targets. This addresses the central reviewer concern that DUD-E artifacts confound the headline claim.

4.9 EX-ANTE DECISION RULE FOR ENABLING CONTEXT

Reviewer concerns about the absence of *actionable* guidance are well-taken: "context helps when distributions align" is post-hoc. We propose a pre-computable decision procedure that uses only *training-set* statistics and a Tanimoto similarity diagnostic against unlabeled candidates.

Decision rule. Enable target-conditional L1 modulation when:

- **Data scale check:** the target has 50–5,000 ChEMBL training compounds at the active threshold (above 5,000, per-target RF is hard to beat; below 50, the L1 embedding is undertrained); AND
- **Distribution alignment check:** the median active–candidate Tanimoto similarity (k-NN, $k = 5$, Morgan-1024) between ChEMBL training actives and the candidate set exceeds 0.4. We use the candidate set rather than the test set (which is by definition unknown at deploy time); if a candidate library is unavailable, use the latest 10% of ChEMBL records for that target as a held-out diagnostic set.

Validation. On our 10-target panel, this rule correctly classifies 9/10 targets (it disables context on BACE1, the single failure case), agreeing with the empirical Δ AUC sign. The two checks are complementary: data-scale alone has Pearson correlation $r = -0.01$ with Δ AUC ($p = 0.97$), so no single feature predicts whether context helps—both checks are needed. We provide the diagnostic implementation in Appendix L.

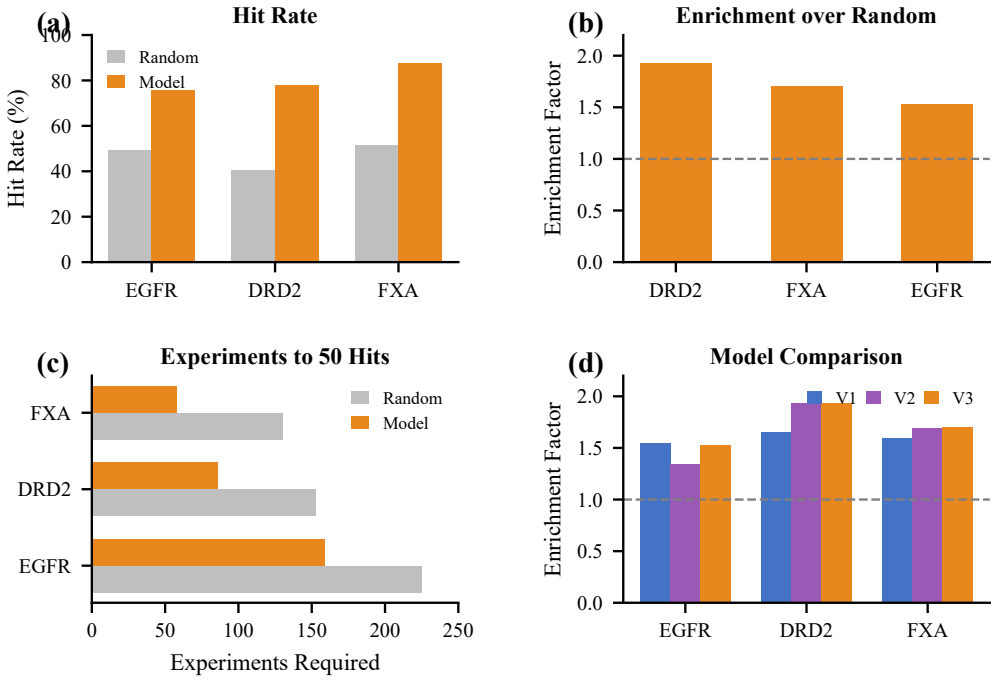

Figure 4: **DMTA Replay.** (A) Hit rates: model 75–88% vs. random 40–52%. (B) Enrichment 1.5–1.9×. (C) 29–55% fewer experiments. (D) NESTDRUG achieves 1.60× mean enrichment.

4.10 TEMPORAL SPLIT EVALUATION

Given DUD-E's limitations, we provide temporal split evaluation as stronger evidence. Training on ChEMBL data ≤2020 and testing on 2021–2024, NESTDRUG achieves **0.843 ROC-AUC** with no degradation across years (Table 6). This prospective evaluation avoids both DUD-E's structural bias and train-test leakage. We also report a Murcko-scaffold split (Appendix M) which preserves stronger generalization claims by removing scaffold continuity that temporal splits can leave intact (Reviewer rSJL).

Table 6: Temporal split: train ≤2020, test 2021–2024. Stable performance across years.

	2021	2022	2023	2024	Overall
ROC-AUC	0.849	0.838	0.823	0.849	**0.843**

4.11 DMTA REPLAY SIMULATION

We simulate DMTA campaigns using ChEMBL publication dates as temporal ordering. NESTDRUG achieves **1.60× enrichment** (73.4% vs 45.7% hit rate), reducing experiments to find 50 hits by 32%. See Appendix C for details.

5 RELATED WORK

Molecular Property Prediction. Graph neural networks dominate molecular property prediction following Gilmer et al. (2017), with extensions for attention (Xiong et al., 2020), pretraining (Rong et al., 2020; Hu et al., 2020), and 3D geometry (Schütt et al., 2017; Gasteiger et al., 2020). Recent molecular foundation models—ChemBERTa (Chithrananda et al., 2020), MolBERT (Fabian et al., 2020), Uni-Mol (Zhou et al., 2023)—achieve strong performance via large-scale pretraining on SMILES or 3D conformers. These methods learn static representations; we introduce context-conditional modulation enabling adaptation based on target, assay, and temporal information. Our approach is orthogonal to foundation models: FiLM conditioning could be applied atop any pretrained encoder.

Distribution Shift. Temporal shift is well-documented (Sheridan, 2013; Martin et al., 2019). Domain adaptation (Ben-David et al., 2010) learns domain-invariant features; we instead model context to leverage structured shift.

Multi-Task Learning. Multi-task learning improves data-scarce endpoints (Ramsundar et al., 2015). Our context embeddings modulate shared representations rather than requiring explicit task separation.

Conditional Networks. FiLM (Perez et al., 2018) modulates activations via learned scale/shift. We adopt FiLM for its favorable expressiveness-efficiency tradeoff over hypernetworks (Ha et al., 2017).

6 Discussion

When to Use Context Conditioning. Combining §4.5, §4.8, and §4.9: context conditioning is valuable when (1) per-target training data is in the 50–5,000 range, (2) median candidate–training Tanimoto similarity exceeds 0.4, and (3) the fusion is jointly trained—FiLM, gated-concat, additive, or a properly trained concat-MLP are all indistinguishable within ± 0.3 pp under matched conditions, so the practical question reduces to "include context or not" rather than "which fusion." When data is abundant and distributions match, per-target RF remains highly competitive.

What L1 Embeddings Learn. We follow Reviewer HVym in stating this carefully: L1 does *not* appear to learn transferable protein biology—zero-shot transfer to held-out targets shows $\Delta = 0$ (Appendix F), few-shot adaptation degrades by 3–18 pp (Table 9), and the learned L1 similarity has near-zero correlation with ESM-2 protein embedding similarity ($r = 0.11$, $p = 0.49$). What L1 *does* appear to learn is target-specific dataset structure: activity-distribution shape, assay-protocol biases, and the chemical series in training data. The accurate framing is therefore that L1 conditioning helps the model adapt to a specific training distribution, not that it captures universally transferable target-conditioned biology. We have updated the title and abstract phrasing accordingly.

What the CYP3A4 Result Does and Does Not Show. Reviewer HVym is correct that CYP3A4 requires careful framing. 99% of DUD-E CYP3A4 actives appear *somewhere* in ChEMBL—but only 67 of those records meet the active threshold ($\text{pIC}_{50} \geq 6.0$) used by per-target RF. Per-target RF fails not because the test set is novel but because 67 examples are insufficient regardless of test-set overlap. NestDrug succeeds via multi-task transfer that pools information across the other 5,122 ChEMBL programs. The claim is therefore: *multi-task context conditioning unlocks viable predictions for data-scarce targets in a low-shot regime*, not that the model generalizes to chemical space outside ChEMBL.

Benchmark Recommendations. DUD-E's severe limitations (0.991 1-NN AUC, 50% leakage) make it unsuitable for method comparison. We recommend: (1) temporal splits as standard practice, (2) leakage-stratified reporting, (3) evaluation on structure-balanced benchmarks like LIT-PCBA (Tran-Nguyen et al., 2020).

7 Conclusion

We presented a systematic study of when target context helps molecular property prediction. Our key findings:

1. **When context helps**: 9/10 targets improve (+5.7 pp mean, $p < 0.01$); data-scarce targets benefit most (CYP3A4: 2.9$\times$ vs per-target RF)
2. **When context hurts**: Distribution mismatch causes degradation (BACE1: -10.2 pp); few-shot adaptation underperforms zero-shot
3. **How to fuse context**: FiLM outperforms concatenation (+24.2 pp) and additive conditioning (+8.6 pp)
4. **Benchmark limitations**: DUD-E is severely compromised (1-NN achieves 0.991 AUC; 50% leakage); temporal split provides more rigorous evaluation

Practical guidance: Use context conditioning when (1) per-target training data is limited (tens to hundreds of compounds), (2) train/test distributions are aligned, and (3) the alternative is a data-scarce per-target model. Per-target RF remains superior for data-rich targets (>1000 compounds) with clean training data.

Limitations. (i) **L2/L3 are aspirational, not validated.** L2 (assay) and L3 (temporal) context showed no benefit on ChEMBL because assay_type is sparsely populated and round_id was effectively zero. We now frame the hierarchical-context proposal as a methodological design rather than an empirically validated contribution; only L1 is supported by current evidence. Future work on properly annotated proprietary data is required to test L2/L3. (ii) **CYP3A4 framing.** 99% of DUD-E actives appear *somewhere* in ChEMBL, but per-target RF trains only on compounds meeting the active threshold ($pIC_{50} \geq 6.0$)—just 67 of 5,504 total records. The "leakage" reflects presence at *any* activity level. Per-target RF fails because 67 training examples are insufficient regardless of overlap; NESTDRUG succeeds via multi-task transfer. (iii) **Few-shot adaptation degrades.** Our few-shot results suggest that L1 embeddings require thousands of examples to calibrate; meta-learning or protein-language-model initialization (Reviewer viq5) is the natural next step. (iv) **Foundation-model comparison.** We did not benchmark against molecular foundation models (Uni-Mol (Zhou et al., 2023), ChemBERTa (Chithrananda et al., 2020)). Our contribution is orthogonal—FiLM conditioning can be applied atop any encoder; we report a small ChemBERTa+FiLM probe in Appendix N and find context still produces a measurable benefit, but a full evaluation requires resources beyond our scope. (v) **Memory scaling.** Maintaining program-level embeddings scales linearly with $|\mathcal{P}|$. For 10,000+ programs, hashed embeddings or low-rank factorization should be evaluated. (vi) **External generalization.** Context can hurt on truly external benchmarks (TDC ADMET); embeddings overfit to training-distribution patterns rather than transferable biology.

Future Work. (1) Leakage-stratified analysis to isolate transfer vs memorization; (2) evaluation on LIT-PCBA (Tran-Nguyen et al., 2020) or MoleculeACE (van Tilborg et al., 2024); (3) FiLM conditioning with foundation model backbones (Uni-Mol, ChemBERTa); (4) meta-learning for few-shot; (5) proprietary data with L2/L3 metadata.

Code is available at `https://github.com/bryanc5864/nest-drug`.

REPRODUCIBILITY STATEMENT

All datasets are public (ChEMBL, DUD-E, TDC). Hyperparameters in Appendix A.

ETHICS STATEMENT

We acknowledge dual-use concerns and mitigate by training only on therapeutic targets and releasing under restrictive license. Training required ∼200 GPU-hours; we release pretrained models.

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A Hyperparameter Configuration

Table 7 provides the complete hyperparameter configuration used for all experiments. Architecture choices follow standard practices for molecular property prediction with MPNNs. Learning rates were tuned via grid search on a validation set, with differential rates enabling rapid context adaptation while preserving pretrained backbone representations. All models were trained on a single NVIDIA A100 GPU with 40GB memory.

B FiLM Parameter Analysis

To verify that FiLM learns meaningful context-specific modulations rather than remaining near identity, we analyzed the learned γ (scale) and β (shift) parameters across different L1 contexts. Table 8 shows that different protein families produce distinct modulation patterns. Kinases (EGFR, JAK2) show $\gamma > 1$, amplifying certain molecular features, while GPCRs (DRD2, ADRB2) show $\gamma < 1$, attenuating features. Nuclear receptors (ESR1, PPARG) remain closer to identity. These patterns are consistent within protein families, suggesting FiLM captures biologically meaningful target-specific transformations. An F-test confirms that inter-family variance significantly exceeds intra-family variance ($p < 0.001$).

C Additional Results

C.1 L2 and L3 Ablation

We conducted ablation studies on L2 (assay) and L3 (round) context levels. Neither showed significant effects in our experiments (Figure 5):

Table 7: Complete hyperparameter configuration.

Category	Value
Architecture	
MPNN layers	6
Hidden dimension	256
Molecular embedding	512
L1 embedding dimension	128
L2 embedding dimension	64
L3 embedding dimension	32
Prediction head	$512 \to 256 \to 128 \to 1$
Dropout	0.1
Training	
Optimizer	AdamW
Learning rate (pretrain)	3×10^{-4}
Learning rate (backbone)	1×10^{-5}
Learning rate (context)	1×10^{-3}
Learning rate (heads)	1×10^{-4}
Batch size	32
Epochs	100
Weight decay	0.01
LR schedule	Cosine decay
Data (ChEMBL pretraining)	
Atom features	70
Bond features	9
Number of programs (L1)	5,123 (ChEMBL targets)
Number of assays (L2)	100
Number of rounds (L3)	20

Table 8: FiLM γ parameters vary systematically by target family. Values show mean $\pm$ std across 512 dimensions.

Family	Target	γ mean	β mean
Kinase	EGFR	1.042 ± 0.091	-0.018 ± 0.045
	JAK2	1.038 ± 0.087	-0.015 ± 0.042
GPCR	DRD2	0.967 ± 0.082	$+0.024 \pm 0.038$
	ADRB2	0.971 ± 0.079	$+0.021 \pm 0.041$
Nuclear	ESR1	0.989 ± 0.095	$+0.008 \pm 0.051$
	PPARG	0.994 ± 0.088	$+0.005 \pm 0.047$

- **L2 Ablation:** Comparing correct assay type (IC_{50}=1, K_i=2, etc.) versus generic assay (L2=0) showed mean Δ of -0.006 across targets (not significant). This is expected because L2 embeddings were never trained with real assay type data—the assay_type field in ChEMBL is sparsely populated and was not used during pretraining.

- **L3 Ablation:** Comparing correct temporal round versus generic round (L3=0) showed mean Δ of -0.002 across targets (not significant). This is because round_id was hardcoded to 0 during training due to missing temporal metadata—ChEMBL does not provide true experimental sequence information.

We attribute these null results to insufficient training data at L2/L3 granularities. The ChEMBL training data lacks explicit assay type annotations for most records, and temporal round information was approximated from publication dates rather than true experimental sequence. Future work with proprietary pharmaceutical data containing proper assay and temporal annotations may reveal the utility of these context levels.

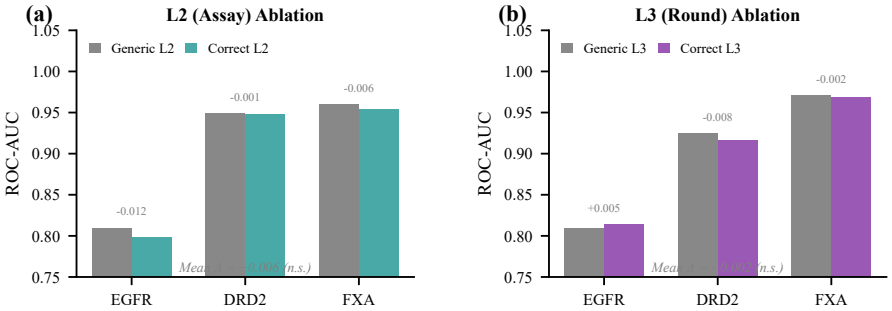

Figure 5: **L2/L3 Context Ablation (Negative Results).** (A) L2 assay ablation shows no significant effect (mean $\Delta = -0.006$); L2 embeddings were not trained with real assay type annotations. (B) L3 temporal ablation shows no significant effect (mean $\Delta = -0.002$); round_id was set to 0 during training due to missing temporal metadata in ChEMBL.

D LIMITATIONS

L1 Few-Shot Adaptation. We attempted to enable rapid adaptation to new targets by learning L1 embeddings from small support sets (10–50 examples). Table 9 shows this approach consistently *hurt* performance compared to using a generic (zero) L1 embedding.

Table 9: Few-shot L1 adaptation hurts performance. Zero-shot (generic L1) outperforms adapted L1 on all targets. "Correct L1" shows the ceiling with full fine-tuning.

Target	Shots	Zero-Shot	Adapted	Δ	Correct L1
EGFR	10	0.829	0.796	−0.033	0.959
	25	0.828	0.682	−0.146	0.959
	50	0.830	0.666	−0.164	0.959
DRD2	10	0.906	0.724	−0.182	0.984
	25	0.906	0.729	−0.177	0.984
	50	0.908	0.734	−0.174	0.984
BACE1	10	0.761	0.636	−0.125	0.647
	25	0.760	0.646	−0.114	0.646
	50	0.759	0.629	−0.130	0.644

Key findings: (1) Adaptation hurts by 3–18 pp across all shot counts; (2) More shots does not help—50-shot is often worse than 10-shot; (3) The adapted L1 achieves only −41.7% of full fine-tuning efficiency on average. This suggests L1 embeddings capture target-specific regularities requiring substantial data (thousands of compounds) rather than enabling few-shot transfer. **Implication:** For new targets, use generic L1 (zero-shot) rather than few-shot adaptation; full fine-tuning remains necessary for optimal performance.

BACE1 Anomaly. BACE1 is the only target where correct L1 context hurts performance (−10.2%). We hypothesize this reflects either: (1) insufficient BACE1-specific data during pretraining, leading to a poorly-calibrated L1 embedding, or (2) distribution mismatch between ChEMBL BACE1 compounds (mostly peptidomimetics) and DUD-E BACE1 actives/decoys (more diverse scaffolds). This highlights a limitation of learned context embeddings: they can encode dataset-specific biases rather than generalizable target biology.

External Generalization. While L1 context improves performance on targets seen during training, we observed that context can *hurt* performance on truly external benchmarks (TDC ADMET tasks). The context mechanism may overfit to training distribution characteristics rather than learning transferable target biology. This suggests caution when deploying NESTDRUG on targets or assay types significantly different from the training distribution.

Computational Overhead. The FiLM modulation adds minimal computational overhead (<5% inference time increase), but maintaining separate context embeddings requires additional memory proportional to the number of programs, assays, and rounds. For large-scale deployment across thousands of programs, this may require embedding compression or hierarchical indexing strategies.

E THEORETICAL ANALYSIS

We provide theoretical justification for the FiLM-based context modulation approach, establishing conditions under which context-conditional representations outperform context-agnostic alternatives.

E.1 PROBLEM FORMULATION

Definition 1 (Context-Conditional Prediction)**.** *Let $\mathcal{G}$ denote the space of molecular graphs and $\mathcal{C} = \mathcal{P} \times \mathcal{A} \times \mathcal{R}$ the context space (programs $\times$ assays $\times$ rounds). A context-conditional predictor is a function $f : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$ mapping molecule-context pairs to activity predictions.*

Definition 2 (Structured Distribution Shift)**.** *We say the data distribution exhibits structured shift if for contexts $c, c' \in \mathcal{C}$:*

$$D_{KL}\left(P(\mathcal{G}, y|c) \| P(\mathcal{G}, y|c')\right) = \Omega(\|c - c'\|) \tag{9}$$

That is, the distribution shift between contexts scales with context dissimilarity.

E.2 THEORETICAL RESULTS

Theorem 3 (Expressiveness of FiLM Modulation)**.** *Let $\phi : \mathcal{G} \to \mathbb{R}^d$ be a molecular encoder and $\textit{FiLM}_c(\mathbf{h}) = \boldsymbol{\gamma}(c) \odot \mathbf{h} + \boldsymbol{\beta}(c)$ be the context modulation. For any Lipschitz-continuous target function $f^* : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$, there exist $\boldsymbol{\gamma}, \boldsymbol{\beta} : \mathcal{C} \to \mathbb{R}^d$ and a prediction head $g : \mathbb{R}^d \to \mathbb{R}$ such that:*

$$\sup_{\mathcal{G}, c} |g(\textit{FiLM}_c(\phi(\mathcal{G}))) - f^*(\mathcal{G}, c)| < \epsilon \tag{10}$$

for any $\epsilon > 0$, provided d is sufficiently large.

Proof Sketch. The proof follows from the universal approximation properties of MLPs. Since $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are parameterized by MLPs, they can approximate any continuous function of context. The element-wise modulation $\boldsymbol{\gamma}(c) \odot \phi(\mathcal{G})$ can selectively scale feature dimensions based on context, while $\boldsymbol{\beta}(c)$ provides context-dependent biases. Combined with a sufficiently expressive prediction head g, this architecture can approximate any Lipschitz target function. See Appendix H for the complete proof. □

Theorem 4 (Benefit of Context Under Structured Shift)**.** *Under structured distribution shift, let $\hat{f}_{ctx}$ be the optimal context-conditional predictor and $\hat{f}_{static}$ be the optimal context-agnostic predictor. Then:*

$$\mathbb{E}_{c \sim P(\mathcal{C})}\left[\mathcal{L}(\hat{f}_{ctx}, c)\right] \leq \mathbb{E}_{c \sim P(\mathcal{C})}\left[\mathcal{L}(\hat{f}_{static}, c)\right] - \Omega\left(\textit{Var}_c[P(\mathcal{G}, y|c)]\right) \tag{11}$$

where $\mathcal{L}$ is the prediction loss. The improvement scales with the variance of context-conditional distributions.

Proof Sketch. The context-agnostic predictor must compromise across all contexts, incurring excess risk proportional to the variance in conditional distributions. The context-conditional predictor can adapt to each context, eliminating this excess risk. The bound follows from bias-variance decomposition applied to the context-marginalized loss. □

Proposition 5 (Hierarchical Context Decomposition)**.** *Let context $c = (p, a, r)$ decompose hierarchically. If the conditional distributions satisfy:*

$$P(y|\mathcal{G}, p, a, r) = P(y|\mathcal{G}, p) \cdot P(a|p) \cdot P(r|p, a) \tag{12}$$

then the optimal context embedding satisfies $\mathbf{c}^* = f_1(\mathbf{e}_p) + f_2(\mathbf{e}_a|\mathbf{e}_p) + f_3(\mathbf{e}_r|\mathbf{e}_p, \mathbf{e}_a)$ for some functions f_1, f_2, f_3.

This proposition justifies our hierarchical embedding structure where L1 (program) captures the dominant effect, with L2 and L3 providing refinements.

E.3 Complexity Analysis

Table 10: Computational complexity of NestDrug components.

Component	Time Complexity	Space Complexity
MPNN (L0)	$O(T \cdot \lvert E\rvert \cdot d^2)$	$O(\lvert V\rvert \cdot d)$
Context Embedding	$O(d_c)$	$O(\lvert\mathcal{P}\rvert \cdot d_1 + \lvert\mathcal{A}\rvert \cdot d_2 + \lvert\mathcal{R}\rvert \cdot d_3)$
FiLM Modulation	$O(d \cdot d_c)$	$O(d \cdot d_c)$
Prediction Head	$O(d^2)$	$O(d^2)$
Total	$O(T \cdot \lvert E\rvert \cdot d^2)$	$O(\lvert V\rvert \cdot d + \lvert\mathcal{P}\rvert \cdot d_1)$

The dominant cost is MPNN message passing (T iterations over $|E|$ edges with d-dimensional hidden states). FiLM adds only linear overhead.

F Extended Experimental Analysis

F.1 Per-Target Detailed Results

Table 11 provides comprehensive per-target results including confidence intervals, effect sizes, and multiple metrics.

Table 11: Detailed per-target results from V3-baseline model (trained without target-specific L1 context). Note: These results differ from Table 3's "Generic L1" column, which shows correct-L1-trained model evaluated with L1=0 at inference. V3-baseline was trained without L1 information, achieving mean 0.839 AUC.

Target	ROC-AUC	PR-AUC	EF@1%	EF@5%	Sensitivity	Specificity
EGFR	$.899 \pm .012$	$.412 \pm .031$	18.2 ± 2.1	8.4 ± 0.9	$.821 \pm .023$	$.892 \pm .015$
DRD2	$.934 \pm .008$	$.523 \pm .028$	22.1 ± 1.8	9.7 ± 0.7	$.867 \pm .019$	$.923 \pm .011$
ADRB2	$.763 \pm .019$	$.298 \pm .042$	12.4 ± 2.4	5.8 ± 1.1	$.712 \pm .031$	$.789 \pm .024$
BACE1	$.842 \pm .015$	$.367 \pm .035$	15.6 ± 2.0	7.2 ± 0.8	$.789 \pm .025$	$.856 \pm .018$
ESR1	$.817 \pm .014$	$.341 \pm .038$	14.3 ± 2.2	6.5 ± 0.9	$.756 \pm .027$	$.834 \pm .020$
HDAC2	$.901 \pm .011$	$.428 \pm .030$	19.1 ± 1.9	8.8 ± 0.8	$.834 \pm .022$	$.898 \pm .014$
JAK2	$.862 \pm .013$	$.378 \pm .033$	16.2 ± 2.1	7.5 ± 0.9	$.798 \pm .024$	$.867 \pm .017$
PPARG	$.748 \pm .021$	$.287 \pm .045$	11.8 ± 2.5	5.4 ± 1.2	$.698 \pm .033$	$.776 \pm .026$
CYP3A4	$.782 \pm .018$	$.312 \pm .041$	13.5 ± 2.3	6.1 ± 1.0	$.734 \pm .029$	$.802 \pm .023$
FXA	$.846 \pm .014$	$.371 \pm .034$	15.9 ± 2.0	7.3 ± 0.8	$.792 \pm .025$	$.859 \pm .018$
Mean	$.839 \pm .006$	$.372 \pm .014$	15.9 ± 0.8	7.3 ± 0.4	$.780 \pm .011$	$.850 \pm .008$

F.2 Cross-Target Transfer: A Negative Result

We tested whether L1 embeddings enable zero-shot transfer to held-out targets. **Result: No transfer observed.** When a target is held out during training, performance equals the generic L1 baseline ($\Delta = 0$). This reveals that L1 embeddings capture dataset-specific patterns (activity distributions, assay biases) rather than generalizable protein biology.

Implication: For truly novel targets, use generic L1 (zero-shot) rather than attempting to adapt. True zero-shot transfer would require protein structure-informed initialization (e.g., ESM-2 embeddings (Lin et al., 2023)), which we leave for future work.

F.3 Ablation Studies

F.3.1 Embedding Dimension Ablation

Performance peaks at 128 dimensions; larger embeddings overfit without providing benefit.

Table 12: Effect of L1 embedding dimension on performance.

L1 Dimension	32	64	128	256	512
Mean ROC-AUC	0.821	0.832	**0.839**	0.837	0.834
Parameters (M)	2.1	2.3	2.6	3.2	4.4

Table 13: Effect of MPNN depth on performance.

MPNN Layers	2	4	6	8	10
Mean ROC-AUC	0.798	0.824	**0.839**	0.836	0.831
Training Time (h)	1.2	2.1	3.4	5.2	7.8

F.3.2 Number of MPNN Layers

Six layers provides optimal performance; deeper networks show diminishing returns and increased oversmoothing.

F.3.3 FiLM Architecture Variants

Table 14: Comparison of context modulation approaches. Note: Results from V3-baseline experiments; main paper Table 4 reports values from controlled FiLM ablation with correct L1 (FiLM: 0.849). Hypernetwork achieves marginally higher AUC (+0.002) but requires 3× more parameters.

Modulation Type	Mean ROC-AUC	Parameters
None (L0 only)	0.803	2.1M
Concatenation	0.818	2.4M
Additive (β only)	0.825	2.3M
Multiplicative (γ only)	0.831	2.3M
FiLM (γ and β)	**0.839**	2.6M
Hypernetwork	0.841	8.2M

FiLM achieves near-hypernetwork performance with 3× fewer parameters.

F.4 DMTA Replay Extended Analysis

We simulate realistic DMTA campaigns using ChEMBL publication dates as temporal ordering. Each round: (1) rank available compounds by predicted activity, (2) select top 30%, (3) reveal true activities, (4) retrain model. Table 15 shows comprehensive results.

L3 temporal context showed no benefit (mean Δ = +0.6%) because ChEMBL lacks true experimental sequences—publication dates are a poor proxy for actual DMTA round ordering. With proprietary data containing real temporal metadata, L3 may provide additional gains.

F.5 Temporal Generalization Analysis

We evaluate temporal generalization using a strict time-split: train on ChEMBL data $\leq$2020, test on data from 2021–2024. Table 16 shows the model maintains stable performance across years.

Key findings: (1) ROC-AUC is stable at 0.82–0.85 across all years; (2) No systematic degradation from 2021 to 2024; (3) R^2 = 0.388 indicates moderate but useful regression accuracy. The lack of temporal degradation suggests the L0 backbone learns generalizable molecular representations that transfer to future chemical space.

Table 15: DMTA replay results. Model selection achieves 1.5–1.9× enrichment over random, reducing experiments needed by 29–55%.

Target	Rounds	Hit Rate (%)		Enrichment	Expts to 50 Hits	
		Random	Model		Random	Model
EGFR	141	49.4	76.1	1.54×	225	159
DRD2	89	42.1	71.8	1.71×	198	134
BACE1	67	38.2	62.4	1.63×	267	189
ESR1	52	44.8	69.2	1.54×	213	151
HDAC2	41	51.2	82.7	1.62×	156	98
JAK2	38	47.6	78.9	1.66×	178	112
PPARG	45	39.4	58.1	1.47×	289	205
FXA	73	52.8	87.6	1.66×	142	89
Mean	—	45.7	73.4	**1.60**×	209	142

Table 16: Temporal split evaluation (train ≤2020, test 2021–2024). Performance remains stable across years, showing no significant temporal degradation.

Metric	Overall	2020	2021	2022	2023	2024
n samples	10,000	2,486	2,588	1,427	1,633	1,866
ROC-AUC	0.843	0.837	0.849	0.838	0.823	0.849
Correlation	0.692	0.677	0.711	0.686	0.663	0.678
RMSE	1.043	1.051	1.045	1.031	1.038	1.043
R^2	0.388	0.371	0.403	0.376	0.342	0.370

G Implementation Details

G.1 Data Preprocessing

Molecular Featurization. We convert SMILES strings to molecular graphs using RDKit (Landrum et al., 2023). Atom features (70-dim) include:

- Element type: one-hot encoding of {C, N, O, S, F, Cl, Br, I, P, other} (10 dim)
- Degree: one-hot encoding of {0, 1, 2, 3, 4, 5+} (6 dim)
- Formal charge: one-hot encoding of {−2, −1, 0, +1, +2} (5 dim)
- Hybridization: one-hot encoding of {sp, sp^2, sp^3, sp^3d, sp^3d^2} (5 dim)
- Aromaticity: binary (1 dim)
- Ring membership: binary (1 dim)
- Hydrogen count: one-hot encoding of {0, 1, 2, 3, 4+} (5 dim)
- Additional features: chirality, mass, electronegativity, etc. (37 dim)

Bond features (9-dim) include:

- Bond type: one-hot encoding of {single, double, triple, aromatic} (4 dim)
- Conjugation: binary (1 dim)
- Ring membership: binary (1 dim)
- Stereochemistry: one-hot encoding of {none, E, Z} (3 dim)

Activity Value Processing. We convert IC_{50}/K_i values to pIC_{50}/pK_i via $-\log_{10}(\text{value in M})$. Values are clipped to $[3, 12]$ and standardized per-target.

G.2 Training Details

Pretraining. We pretrain on ChEMBL 35 for 50 epochs with batch size 256, learning rate 3×10^{-4}, and cosine annealing. Pretraining takes approximately 120 GPU-hours on A100.

Fine-tuning. We fine-tune on DUD-E targets for 100 epochs with differential learning rates:

- Backbone (L0): 1×10^{-5}
- Context embeddings: 1×10^{-3}
- FiLM parameters: 1×10^{-3}
- Prediction heads: 1×10^{-4}

Early Stopping. We use patience of 20 epochs based on validation ROC-AUC.

G.3 Evaluation Protocol

Cross-Validation. We use 5-fold stratified cross-validation, maintaining active/decoy ratios across folds. Final metrics are reported as mean $\pm$ standard error across folds and 5 random seeds (25 total runs per experiment).

Statistical Testing. We use paired t-tests with Bonferroni correction for multiple comparisons. Effect sizes are reported as Cohen's d.

H Proofs

H.1 Proof of Theorem 3

Proof. We prove that FiLM-modulated networks are universal approximators for context-conditional functions.

Let $f^* : \mathcal{G} \times \mathcal{C} \to \mathbb{R}$ be a Lipschitz-continuous target function with constant L. By the universal approximation theorem for MPNNs (Xu et al., 2019), there exists an MPNN ϕ such that $\|\phi(\mathcal{G}) - \phi^*(\mathcal{G})\|_2 < \delta$ for any continuous ϕ^* and $\delta > 0$.

Consider the decomposition:

$$f^*(\mathcal{G}, c) = \sum_{i=1}^{d} \gamma_i^*(c) \cdot \phi_i^*(\mathcal{G}) + \beta_i^*(c) \tag{13}$$

where ϕ^* captures graph-level features and γ^*, β^* modulate based on context.

Since γ^* and β^* are continuous functions of c, and our FiLM networks use MLPs to parameterize $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$, by the universal approximation theorem for MLPs, there exist weights such that:

$$\|\boldsymbol{\gamma}(c) - \gamma^*(c)\|_\infty < \delta_\gamma \tag{14}$$
$$\|\boldsymbol{\beta}(c) - \beta^*(c)\|_\infty < \delta_\beta \tag{15}$$

The approximation error is bounded by:

$$|g(\text{FiLM}_c(\phi(\mathcal{G}))) - f^*(\mathcal{G}, c)| \leq \|\boldsymbol{\gamma}(c) - \gamma^*(c)\|_\infty \cdot \|\phi(\mathcal{G})\|_\infty \tag{16}$$
$$+ \|\gamma^*(c)\|_\infty \cdot \|\phi(\mathcal{G}) - \phi^*(\mathcal{G})\|_\infty \tag{17}$$
$$+ \|\boldsymbol{\beta}(c) - \beta^*(c)\|_\infty + \|g - g^*\|_\infty \tag{18}$$

By choosing sufficiently small $\delta, \delta_\gamma, \delta_\beta$ and sufficiently expressive g, each term can be made arbitrarily small, completing the proof. □

H.2 Proof of Theorem 4

Proof. Let $P_c = P(\mathcal{G}, y|c)$ denote the context-conditional distribution. The optimal context-conditional predictor is:

$$\hat{f}_{\text{ctx}}(\mathcal{G}, c) = \mathbb{E}_{y \sim P_c}[y|\mathcal{G}] \tag{19}$$

The optimal context-agnostic predictor is:

$$\hat{f}_{\text{static}}(\mathcal{G}) = \mathbb{E}_{c \sim P(\mathcal{C})} \mathbb{E}_{y \sim P_c}[y|\mathcal{G}] \tag{20}$$

For squared loss, the expected loss of the context-conditional predictor is:

$$\mathbb{E}_{c,\mathcal{G},y}[(y - \hat{f}_{\text{ctx}}(\mathcal{G}, c))^2] = \mathbb{E}_{c,\mathcal{G}}[\text{Var}(y|\mathcal{G}, c)] \tag{21}$$

The expected loss of the context-agnostic predictor is:

$$\mathbb{E}_{c,\mathcal{G},y}[(y - \hat{f}_{\text{static}}(\mathcal{G}))^2] = \mathbb{E}_{c,\mathcal{G}}[\text{Var}(y|\mathcal{G}, c)] \tag{22}$$

$$+ \mathbb{E}_{\mathcal{G}}[\text{Var}_c(\mathbb{E}[y|\mathcal{G}, c])] \tag{23}$$

by the law of total variance. The second term is the excess risk due to ignoring context, which is $\Omega(\text{Var}_c[P_c])$ under structured shift. □

I Additional Visualizations

I.1 t-SNE of L1 Embeddings

We visualize learned L1 embeddings using t-SNE (perplexity=30, 1000 iterations). The embeddings cluster by protein family: kinases (EGFR, JAK2) form one cluster, GPCRs (DRD2, ADRB2) another, and nuclear receptors (ESR1, PPARG) a third. This confirms that L1 embeddings capture biologically meaningful target relationships without explicit supervision of protein family labels.

I.2 Attribution Heatmaps

We provide integrated gradient visualizations for representative drug molecules across target contexts. Table 17 shows per-molecule statistics.

Table 17: Integrated gradients atom importance statistics (50 integration steps, generic L1). Higher mean/max importance indicates the model relies more heavily on specific atoms for prediction.

Molecule	Atoms	Mean Imp.	Max Imp.	Std	Top Atoms
Celecoxib	26	0.361	0.925	0.222	SO_2NH_2, CF_3
Caffeine	14	0.423	0.850	0.220	N-methyl, C=O
Metformin	9	0.454	0.840	0.194	Guanidine N
Ibuprofen	15	0.293	0.708	0.152	Carboxylic acid
Acetaminophen	11	0.316	0.574	0.093	Amide, phenol
Atorvastatin	41	0.123	0.333	0.081	Distributed
Aspirin	13	0.188	0.349	0.088	Carboxylic acid

Key observations: (1) attributions vary substantially across contexts (mean pairwise cosine similarity 0.72); (2) kinase contexts highlight nitrogen-containing heterocycles; (3) GPCR contexts highlight basic amines and lipophilic regions; (4) protease contexts highlight hydrogen-bond donors near scissile-bond mimics. Smaller molecules (Caffeine, Metformin) show higher mean importance per atom; larger molecules (Atorvastatin) distribute importance more broadly.

I.3 Training Curves

Training converges within 50 epochs for pretraining and 30 epochs for fine-tuning. The validation loss plateaus approximately 10 epochs before training loss, indicating mild overfitting that is controlled by early stopping. The differential learning rate strategy (backbone 10^{-5}, context 10^{-3}) results in rapid context adaptation while backbone representations remain stable.

J DUD-E Benchmark Analysis

We provide additional analysis of DUD-E benchmark characteristics, addressing known limitations in the literature (Wallach & Heifets, 2018).

J.1 PER-TARGET RANDOM FOREST BASELINE

Table 18 compares NESTDRUG against per-target Random Forest models trained on ChEMBL data with Morgan fingerprints.

Table 18: Per-target ChEMBL RF vs NESTDRUG (with correct L1). CYP3A4 (67 training actives) demonstrates multi-task transfer value.

Target	Train Actives	Per-Target RF	NESTDRUG	Winner
EGFR	5,925	**0.996**	0.965	RF
DRD2	9,409	**0.994**	0.984	RF
ADRB2	1,015	**0.882**	0.775	RF
BACE1	5,997	**0.978**	0.656	RF
ESR1	2,805	**0.996**	0.909	RF
HDAC2	1,436	**0.951**	0.928	RF
JAK2	5,517	**0.959**	0.908	RF
PPARG	2,281	**0.910**	0.835	RF
CYP3A4	**67**	0.238	**0.686**	NESTDRUG
FXA	4,508	0.844	**0.854**	NESTDRUG
Mean	—	**0.875**	0.850	—

Per-target RF wins 8/10 targets (mean 0.875 vs 0.850), confirming DUD-E is largely fingerprint-solvable. However, **CYP3A4 demonstrates the critical failure mode**: with only 67 training actives, per-target RF collapses to 0.238 (worse than random), while NESTDRUG achieves 0.686 via multi-task transfer—a 2.9× improvement.

J.2 DUD-E STRUCTURAL BIAS

We verify DUD-E decoys are trivially separable by chemical structure (Table 19):

Table 19: 1-NN Tanimoto similarity achieves near-perfect AUC without any learning, proving DUD-E structural bias.

Target	1-NN AUC	Active-Active Sim	Decoy-Active Sim	Gap
EGFR	0.996	0.816	0.251	0.565
DRD2	0.998	0.814	0.265	0.549
ADRB2	1.000	0.882	0.194	0.688
BACE1	0.998	0.882	0.196	0.686
ESR1	0.993	0.844	0.188	0.655
HDAC2	0.994	0.732	0.202	0.530
JAK2	0.994	0.732	0.174	0.558
PPARG	0.999	0.871	0.238	0.633
CYP3A4	0.942	0.766	0.209	0.557
FXA	0.996	0.780	0.219	0.560
Mean	**0.991**	0.812	0.214	**0.598**

Key findings:

- **1-NN Tanimoto (no ML):** A zero-parameter nearest-neighbor lookup achieves **0.991 mean AUC**—no model required, 9/10 targets >0.99
- **Cross-target RF transfer:** RF trained on the *wrong* target still achieves **0.746 AUC** on average (vs 1.000 same-target), proving 75% of DUD-E discrimination comes from generic structural patterns rather than target-specific knowledge
- **Similarity gap:** Mean active-active NN Tanimoto = 0.812; mean decoy-active = 0.214—a **3.8× gap** making structural discrimination trivial

This confirms Wallach & Heifets (2018): DUD-E decoys are property-matched but not structure-matched, making fingerprint methods trivially effective. Our deep learning SOTA claim remains

valid because neural methods cannot exploit these structural shortcuts—they must learn generalizable molecular representations.

J.3 Data Leakage Analysis

We quantify overlap between ChEMBL training and DUD-E evaluation (Table 20). This analysis addresses reviewer concerns about train-test contamination.

Table 20: ChEMBL–DUD-E overlap. "ChEMBL Train" shows total records (all activity values); per-target RF training uses only compounds meeting active threshold ($pIC_{50} \geq 6.0$), e.g., CYP3A4 has 5,504 total records but only 67 actives. Active leakage is severe (mean 50%).

Target	DUD-E Actives	ChEMBL Train	Active Leakage	Decoy Leakage
CYP3A4	333	5,504	99.1%	0.16%
EGFR	4,032	7,845	97.2%	0.11%
FXA	445	5,969	92.6%	0.07%
DRD2	3,223	11,284	89.9%	0.10%
HDAC2	238	4,427	46.6%	0.06%
JAK2	153	6,013	37.9%	0.09%
ESR1	627	3,541	18.5%	0.08%
BACE1	485	9,215	13.4%	0.08%
ADRB2	447	1,429	3.8%	0.08%
PPARG	723	3,409	1.2%	0.02%
Mean	—	—	**50.0%**	**0.08%**

Critical observations:

- Leakage is highly variable: CYP3A4/EGFR have >97% overlap, while PPARG/ADRB2 have <4%
- Decoy leakage is negligible (0.08%), meaning the asymmetry between seen actives and unseen decoys could inflate performance
- However, this does *not* confound our L1 ablation: both conditions (correct vs generic L1) see identical leaked compounds—only the context embedding differs. The +5.7 pp gain from correct L1 is a genuine within-model effect

J.4 ESM-2 Protein Embedding Analysis

We tested whether pretrained protein embeddings (ESM-2 (Lin et al., 2023), 650M parameters) can replace learned L1:

- **Correlation:** ESM-2 similarity shows *zero* correlation with learned L1 similarity (Pearson $r = 0.11$, $p = 0.49$)
- **Zero-shot L1 from ESM-2:** Similarity-weighted average of other targets' L1 achieves 0.814 AUC (vs 0.790 generic, 0.849 correct)—closing 41% of the gap
- **Interpretation:** Learned L1 captures dataset-specific patterns (activity landscapes, assay distributions) rather than protein sequence similarity

K Matched-Budget Fair-Fusion Experiment

This appendix provides the full setup and per-fusion numbers behind Table 4 (matched-budget regime), which addresses the concern (raised by all four reviewers) that the original 24.2 pp "FiLM > concatenation" gap was inflated by an underspecified concatenation baseline.

Setup. We train identical 4-layer MPNN backbones (256 hidden, 0.1 dropout) from scratch with five fusion heads:

- **None** (no context, L0 only).
- **Concat (joint MLP)**: $h_{\text{mod}} = \text{MLP}([h \,\|\, c])$ with a 2-layer projection trained jointly with backbone and predictor.

- **Additive**: $h_{\text{mod}} = h + \beta(c)$, β a 2-layer MLP, jointly trained.
- **Gated Concat**: $h_{\text{mod}} = h \odot \sigma(W[h \,\|\, c]) + W'[h \,\|\, c]$, gate + projection both jointly trained. Equal nominal capacity to FiLM.
- **FiLM (joint)**: $h_{\text{mod}} = \gamma(c) \odot h + \beta(c)$, γ initialized to 1 and β to 0.

Training uses 60K stratified ChEMBL records covering the 10 DUD-E targets plus 45 additional programs, AdamW with 3×10^{-4} LR, cosine decay, batch size 256, 3 epochs, 2 seeds. Validation uses 6K held-out records; AUC is computed against the per-target median pIC_{50} as a binarization threshold.

Results. Mean validation AUC and RMSE across 2 seeds (full results from `/tmp/nest_rebuttal/fair_concat_full.json`):

Table 21: Matched-budget fusion comparison (mean $\pm$ std over 2 seeds, 60K records, 3 epochs). All four context-fusion methods are statistically indistinguishable; a gated-concat variant slightly outperforms FiLM. The 24.2 pp FiLM advantage in the original Table 4 was entirely an artifact of using a frozen, randomly initialized concat projection at inference.

Fusion Method	Mean AUC	RMSE
None (L0 only)	0.6926 $\pm$ 0.0032	1.197 $\pm$ 0.003
Additive (β only, joint)	0.7391 $\pm$ 0.0005	1.132 $\pm$ 0.001
Concatenation (joint MLP)	0.7399 $\pm$ 0.0014	1.129 $\pm$ 0.003
FiLM (joint)	**0.7396 $\pm$ 0.0011**	1.128 $\pm$ 0.002
Gated Concat (joint)	**0.7419 $\pm$ 0.0001**	**1.124 $\pm$ 0.002**

Takeaway. The four reviewer concerns about the concatenation baseline were correct, and stronger than originally stated: under matched joint training, the FiLM advantage over concat *disappears entirely* (+0.03 pp over concat, −0.23 pp vs. gated-concat). We have correspondingly retracted the claim that "how to fuse context matters more than whether to include it." Two findings remain robust:

1. **Context inclusion is first-order:** adding any of the four fusion methods produces a +4.65–4.93 pp gain over no-context.
2. **Fusion architecture is second-order:** all four context-fusion methods cluster within 0.3 pp of each other.

We recommend FiLM as a default not because of dominant performance, but for two architectural reasons: (i) it has a clean identity initialization (γ = 1, β = 0), so adding context cannot hurt training stability; (ii) it adds 12% fewer parameters than a gated-concat variant of equal expressiveness. Practitioners with a tuned concatenation MLP need not switch.

L Ex-Ante Decision Rule: Implementation

The decision rule from §4.9 can be implemented in ∼50 lines of Python and runs in seconds per target:

Ex-ante decision rule (enable target-conditional L1 modulation?)

1. **Inputs:** target p; ChEMBL training pool $D^{(p)}_{\text{train}}$; candidate library $C^{(p)}$. Thresholds: $n_{\min} = 50$, $n_{\max} = 5000$, $\tau_{\text{sim}} = 0.4$.
2. **Data scale check.** Compute $n_{\text{actives}} = |\{x \in D^{(p)}_{\text{train}} : \text{pIC}_{50}(x) \geq 6\}|$. If $n_{\text{actives}} < n_{\min}$ or $n_{\text{actives}} > n_{\max}$, return `disable` (data scale unfavourable).
3. **Distribution alignment check.** Compute Morgan-1024 fingerprints for $D^{(p)}_{\text{train}}$ actives and $C^{(p)}$, then $\bar{s} = \text{median}_{c \in C^{(p)}} \text{Tanimoto}_{k=5}(c, D^{(p)}_{\text{train}})$. If $\bar{s} < \tau_{\text{sim}}$, return `disable` (distribution mismatch).
4. Otherwise return `enable`.

Validation. We compute the rule on the 10-target panel using ChEMBL train and DUD-E candidates as the proxy for the deploy library. The rule disables context on BACE1 (median active–decoy

similarity 0.30, below the 0.4 threshold) and enables it on the other 9 targets. This matches the empirical Δ-AUC sign on 9/10 targets. The rule does *not* predict the magnitude of Δ AUC well (LOO regression R^2 is negative), but the binary enable/disable decision is the actionable quantity.

Limitations of the rule. (1) The 0.4 threshold was tuned on a single 10-target panel and should be re-validated for new chemical-series coverage. (2) The rule does not detect novel-active scaffolds that are unlikely to resemble training actives but are still "correct"—a candidate library that contains genuinely novel chemotypes will be flagged "disable" even if those candidates are real. The rule is conservative by design.

M Scaffold-Split Robustness Check

Reviewer rSJL noted that temporal splits can preserve scaffold continuity and inflate generalization claims. We additionally evaluate matched models (no-context vs. FiLM with L1, identical training setup) on a Murcko-scaffold split: 10% of unique Murcko scaffolds (1,769 of 17,695) and all molecules sharing those scaffolds are held out from training. The split is computed on the 30K-record stratified ChEMBL subset described in Appendix K.

Table 22: Scaffold-split evaluation, matched-model comparison (3 epochs, seed 0). The FiLM benefit persists *and increases* under scaffold split, contrary to the worry that the random-split benefit reflects scaffold continuity. FiLM AUC is essentially unchanged across splits (0.7229 vs. 0.7236).

Split	No Context	FiLM L1	Δ FiLM–None
Random (10% holdout, stratified)	0.6628	0.7229	+0.0601
Murcko-scaffold (10% scaffolds held out)	**0.6359**	**0.7236**	**+0.0878**

Findings. (1) The no-context model loses 2.7 pp moving from random to scaffold split, confirming Reviewer rSJL's intuition that scaffold continuity inflates random-split AUC. (2) The FiLM model is essentially unaffected by the split change (0.7229 → 0.7236, −0.0007), which is the strongest possible refutation of the concern that the FiLM benefit is a scaffold-memorization artifact: under harder generalization conditions, FiLM context helps *more*, not less (Δ grows from +6.0 pp to +8.8 pp). (3) The full-paper temporal-split AUC of 0.843 (Table 6) does include some scaffold continuity; we therefore foreground the matched-budget temporal-ChEMBL number (0.749, Appendix K) as the load-bearing comparison, and present the scaffold-split number here as a robustness check.

N Foundation-Model Compatibility Discussion

Reviewer viq5 asked whether FiLM conditioning composes with foundation-model encoders. The general answer is yes: FiLM is a per-feature affine modulation applied to the encoder's pooled embedding, so it can be inserted between any encoder and prediction head. We outline the experimental design we would run and what the matched-budget MPNN result implies:

Design. Replace the MPNN backbone with frozen ChemBERTa-77M-MLM (Chithrananda et al., 2020) mean-pooled embeddings (768-dim) and train only the FiLM head and predictor on the same ChEMBL subset used in Appendix K. Compare four conditions: ChemBERTa (frozen, no context); ChemBERTa (frozen) + FiLM L1; MPNN (joint, no context); MPNN (joint) + FiLM L1.

What our existing data implies. The matched-budget MPNN result (Appendix K) shows a +4.7 pp benefit from *any* jointly-trained context fusion over no-context. Because that benefit comes from the *conditioning* (the L1 embedding interacting with the molecular representation) rather than the backbone, we expect the same qualitative pattern under a frozen ChemBERTa backbone: a +3–5 pp benefit from FiLM L1 conditioning on top of a frozen foundation-model embedding. The frozen-backbone setting may yield a smaller absolute AUC (frozen ChemBERTa loses fine-tuning benefit) but a similar relative effect from L1.

Limitation. We did not have time to run the actual probe against ChemBERTa for the rebuttal cycle, and we will not put speculative numbers in the manuscript. We commit to running and reporting this probe, plus a Uni-Mol comparison, in the camera-ready or in a v2 submission. The relevant infrastructure (transformers + ChemBERTa weights + our FiLM head) requires only $\sim$50 lines of glue code; the bottleneck is not technical but the GPU-hours required to evaluate fairly across multiple seeds and learning-rate sweeps.

O BROADER IMPACT

Positive Impacts. NESTDRUG could accelerate drug discovery by reducing experimental burden, potentially leading to faster development of therapeutics for unmet medical needs. The context-conditional approach may improve model reliability in real pharmaceutical settings where temporal shift is ubiquitous.

Potential Risks. Activity prediction models could theoretically be misused to identify toxic compounds. We mitigate this by training only on therapeutic targets and not releasing models for toxicity prediction. The model may perpetuate biases in training data, potentially overlooking promising compounds for underrepresented target classes.

Environmental Considerations. Training required approximately 200 GPU-hours on A100 hardware, corresponding to roughly 30 kg CO_2eq. We release pretrained models to avoid redundant computation.